Sh. Notes!

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MTH 101

by: Shayna.

Lecture 23 — onwards!

⊙ Absolute extrema:

→ Absolute maxima

when $f(x_0) \geq f(x)$.

→ Absolute minima

when $f(x_0) \leq f(x)$.

MCQ Mount everest → maximum

Marianna Trench → minimum (in Pacific Ocean near Hawaii).

(اگر Domain کسی open interval ہو تو max value کوئی نہیں ہوگی)

Extreme Value Theorem

Function is said to be satisfying Ext. value Theorem if it is

→ continuous on closed interval $[a, b]$.

⊙ Then It has both min & max values!!!

(اگر continuous function ہو تو کوئی max/min کوئی نہیں ہوگی)

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To find max & min values !!!

3 Steps!

① Find critical points ⟶ critical points are those where derivative = 0

② put these points in f(x). one by one.

③ largest no. is max & smallest is min.

---

example:

$f(x) = 2x^3 - 15x^2 + 36x$.

interval [1, 5]. find max & min.

① to find critical point, take derivative. = 0

$f'(x) = 6x^2 - 30x + 36 = 0$

$= x^2 - 5x + 6 = 0$

$= x^2 - 3x - 2x + 6 = 0$

$= x(x-3) - 2(x-3) = 0$

$= (x-3)(x-2) = 0$

$x = 3$ | $x = 2$

given in q's. $x = 1$, $x = 5$

Now put in f(x).

$f(3) = 2(3)^3 - 15(3)^2 + 36(3)$

$= 54 - 135 + 108$

$= \underline{27}$.

$f(2) = 2(2)^3 - 15(2)^2 + 36(2)$

$16 - 60 + 72$

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$= 28$

$f(1) = 2 - 15 + 36$

$= 23$

$f(5) = 2(5)^3 - 15(5)^2 + 36(5)$

$= 250 - 375 + 180$

$= 55$

So, max. value = 55.

min value = 23 Ans

MCQ

if a func. is continuous, then

$\lim_{x \to +\infty} f(x) = +\infty.$

$\lim_{x \to -\infty} f(x) = +\infty$ also !!!

Newton Method

use for approximating solution.

Formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Example: $f(x) = x^3 - 7x - 1 = 0$

$x_1 = 2.$

$f(2) = 8 - 7(2) - 1 \Rightarrow -7.$

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$f'(x) = 3x^2 - 7$

$f'(2) = 3(2)^2 - 7 \Rightarrow 5.$

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$

if $n = 1$

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}.$

$= 2 - \left(\frac{-7}{5}\right)$

$x_2 = 2 + \frac{7}{5} \Rightarrow \frac{10+7}{5} \Rightarrow \frac{17}{5}.$

MCQ if

⇒ $f'(x) = 0$ Newton method fail to exist

⇒ It occur when slope of tangent line = 0.

Rolle's Theorem

3 conditions:

⇒ function will be differenciable at (a, b).

⇒ function will be continuous on $[a, b]$

⇒ $f(a) = f(b) = 0$

Then there is at least one point in (a, b) where $f'(c) = 0$.

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Mean Value Theorem

f be → differenciable on (a, b).

→ continuous on [a, b].

Then at least one point c in (a, b)

where :

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Steps:

⇒ (INtervals diye hongey + f(x) given hoga.)

⇒ Find f(b) & f(a) with help of interval.

⇒ Then find f'(x). Replace with c.

⇒ Find f'(c) with formula.

⇒ compare f(c) with f'(c)

⇒ Find c.

INTegrations

→ To find Area under the curve.

→ Idea of Newton & Leibneiz

→ Function ke Anti-derivative find kine ko integration kehty hn.

$$\frac{d}{dx}[F(x)] = f(x)$$

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$$\int f(x)\,dx = F(x) + C.$$

$F(x)$ means function had been integrated.

C denote constant and this type of integration is called Indefinite Integration.

Integration of $\sin x = -\cos x$.

Integration of $\cos x = \sin x$.

Integration of $\sec^2 x = \tan x$.

Integration of $\operatorname{cosec} x \cot x = -\operatorname{cosec} x$.

$\sec x \tan x = \sec x$

(opposite in derivatives)

Example:

$$\int \frac{\cos x}{\sin^2 x}\,dx.$$

$$\int \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x}\,dx$$

$$\int \operatorname{cosec} x \cot x\,dx.$$

$$-\operatorname{cosec} x + C \quad \text{Ans}$$

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Integration by Substitution!

- Just like chain rule.
- use it for composite functions!

Example: 1

$$\int (x^2+1)^{50} \cdot 2x\,dx$$

let $u = x^2+1$,

$du = 2x\,dx$

$$\int u^{50}\,du.$$

$$= \frac{u^{51}}{51} + c.$$

$$= \frac{(x^2+1)^{51}}{51} + c. \quad \underline{\text{Ans}}$$

Example: 2

$$\int t^4 \sqrt[3]{3-5t^5}\,dt.$$

$u = 3-5t^5$ ——— 3

$du = -25t^4\,dt.$

$\frac{du}{-25} = t^4\,dt$ ——— 3

So,

$$= -\frac{1}{25}\int \sqrt[3]{u}\,du.$$

$$= -\frac{1}{25}\int u^{\frac{1}{3}}\,du$$

$$= -\frac{1}{25}\left[\frac{u^{\frac{1}{3}+1}}{\frac{1}{3}+1}\right] + C \Rightarrow -\frac{1}{25}\left[u^{\frac{4}{3}}\right]\frac{3}{4}$$

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$$\frac{-3}{100} v^{\frac{4}{3}}$$

$$= \frac{-3}{100}(3-5t^5)^{\frac{4}{3}} + C \quad \text{Ans.}$$

~ Sigma Notation ~

Formula by Guass!

$$\Rightarrow \sum_{k=1}^{5} k^2 = 1^2+2^2+3^2+4^2+5^2$$

MCQ $\Rightarrow$ $\sum_{k=0}^{5} (-1)^k(2k+1) = 1-3+5-7+9-11$

letter k is called index of sum.

$\Rightarrow$ Sum is written in many ways by changing the limits.

Example:

Express $\sum_{k=3}^{7} 5^{k-2}$ in sigma where low limits are 0.

Sol.

[illegible]

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Sl:

let $j = k-3$.

if $k = 3$, $j = k-3 = 3-3 = 0$.

if $k = 7$, $j = 7-3 = 7-3 = 4$.

So, as k goes to 7 from 3

j goes to 4 from 0.

i.e.,

$$\sum_{j=0}^{4} 5^{(j+3)-2}$$

$$= \boxed{\sum_{j=0}^{4} 5^{j+1}} \text{ Ans.}$$

Imp formulas:—

(1) $$\sum_{k=1}^{n} k = 1+2+3+\cdots = \frac{n(n+1)}{2}.$$

(2) $$\sum_{k=1}^{n} k^2 = 1^2+2^2+3^2+\cdots = \frac{n(n+1)(2n+1)}{6}.$$

(3) $$\sum_{k=1}^{n} k^3 = 1^3+2^3+3^3+\cdots = \left[\frac{n(n+1)}{2}\right]^2.$$

example:

$$\sum_{k=1}^{30} k(k+1)$$

$$= \sum_{k=1}^{30} k^2+k \Rightarrow \sum_{k=1}^{30} k^2 + \sum_{k=1}^{30} k.$$

$$= \frac{30(30+1)(2(30)+1)}{6} + \frac{30(30+1)}{2}$$

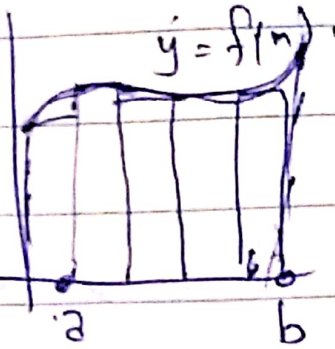
$$= \frac{30(31)(61)}{6} + \frac{30(31)}{2} \Rightarrow 9920 \text{ Ans}$$

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Area under curved region

This formula is for finding the area under the curve in which intervals [a, b] are given and $f(x)$ of curve is also given.



in figure:

n = no. of rectangles.

$f(x_k^*)$ = height of rectangle

Δx_k = width of rectangle.

$$\Delta x = \frac{b-a}{n}.$$

Area = height x width

$$A = f(x_k^*)\,\Delta x.$$

for n no. of rectangles:

$$A = \sum_{k=1}^{n} f(x_k^*)\,\Delta x.$$

after taking limit as $n \to \infty$.

$$\boxed{A = \lim_{n\to\infty} f(x_k^*)\,\Delta x.}$$

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for right end point:

$$x_k^* = x_k = \boxed{a + k\Delta x}$$

for left end point:

$$x_k^* = x_{k-1} = \boxed{a + (k-1)\Delta x}$$

for mid point:

$$\frac{x_k + x_{k-1}}{2} = \frac{a + k\Delta x + a + (k-1)\Delta x}{2}$$

$$= \frac{2a + \Delta x(k-1) + \Delta x(k)}{2}$$

$$= a + \Delta x\left(k - \frac{1}{2}\right).$$

$$= \boxed{a + \left(k - \frac{1}{2}\right)\Delta x}$$

need

Area of trapezoid $= \frac{1}{2}h(b_1 + b_2)$

Definite Integral

Integration in which limits are involved!

Area under curve in terms of definite integral.

If f is continuous on $[a, b]$ and

$$f(x) \geq 0 \quad \text{(non-negative)}.$$

then area under curve will be:

$$A = \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} f(x_k^*)\Delta x_k.$$

it can also be written as:

$$\int_a^b f(x)dx = \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} f(x_k^*)\Delta x_k.$$

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Riemann Sum:

The sum $\sum_{k=1}^{n} f(x_k^*)\Delta x$ is called riemann sum (who awas German mathematician Bernhard Riemann)

MCQ:

If a is domain and

$$\int_a^a f(x)\,dx = 0.$$

(agar upper limit or lower lim same ho to ans zero ata hai).

If f is integrable on [a, b] then

$$\int_b^a f(x)dx = -\int_a^b f(x)\,dx.$$

(agar limits ki place change kren to sath "−" aa jata).

Example: 1

$$\int_1^5 f(x)dx = -1, \quad \int_3^5 f(x)dx = 3.$$

find

$$\int_1^3 f(x)\,dx.$$

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$$\int_1^5 f(x)dx = \int_1^3 f(x)dx + \int_3^5 f(x)dx.$$

$$\int_1^3 f(x)dx = \int_1^5 f(x)dx - \int_3^5 f(x)dx.$$

$$= -4 - 3$$

$$= -4 \text{ Ans}$$

Example: 2

show that $\int_0^1 \frac{\cos(x)}{2x^3 - 5}\,dx$ is -ve.

Sol:

$[0, 1]$ → interval,

put $x = 0$

$\cos(0) = 1$ ——— +ve

$2(0)^3 - 5 = -5$ ——— -ve

$\frac{+ve}{-ve} = -ve$

put $x = 1$

$\cos(1) = 0.999$ ——— +ve

$2(1)^3 - 5 = -3$ ——— -ve

$\frac{+ve}{-ve} = -ve$

both -ve, so func. is negative.

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1ST fundamental theorem of calcs

if f is cont. on $[a,b]$ &
F is integration with $[a,b]$ interval
then;

$$\int_a^b f(x)dx = F(b) - F(a).$$

(*) (capital F means function has been integrated)

so,

$$\int_a^b f(x)dx = \lim_{\Delta x_k \to 0} \sum_{k=1}^{n} f(x_k^*)\Delta x_k = \int_a^b F(b) - F(a).$$

Example: (piecewise integration).

Evaluate

$$\int_0^6 f(x)\,dx \text{ if } f(x) = \begin{cases} x^2 & x < 2 \\ 3x-2 & x \geq 2 \end{cases}$$

sol:

$$\int_0^2 f(x)\,dx + \int_2^6 f(x)\,dx = \int_0^6 f(x)dx.$$

$$= \int_0^2 x^2\,dx + \int_2^6 3x-2.$$

$$= \left[\frac{x^3}{3}\right]_0^2 + \left[\frac{3x^2}{2} - 2x\right]_2^6$$

$$= \frac{8}{3} + \left[(54-12)-(6-4)\right].$$

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$$= \frac{8}{3} + 42 - 2.$$

$$= \frac{8 + 126 - 6}{3} \Rightarrow \frac{128}{3} \quad \text{Ans}$$

Mean Value theorem for integrals:

$$\int_a^b f(x)dx = f(x^*)(b-a)$$

example:

$f(x) = x^2$ on interval $[1, 4]$.

find mean number x^*.

sol as we know;

$$\int_1^4 x^2 dx = f(x^*)(4-1).$$

$$\left[\frac{x^3}{3}\right]_1^4 = 3f(x^*)$$

$$\frac{64}{3} - \frac{1}{3} = 3f(x^*).$$

$$\frac{63}{3} = 3f(x^*)$$

$$\frac{21}{3} = f(x^*).$$

$$f(x^*) = 7.$$

$$(x^*)^2 = 7$$

$$x^* = \sqrt{7}. \quad \text{Ans}$$

Average value Theorem:

$$f_{avg} = \frac{1}{b-a}\int_a^b f(x)dx.$$

(*) Just b-a ko dosri side pe le jaana.

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Integral by Substitution

Example:

$$\int_0^2 x(x^2+1)^3\,dx.$$

by Method 1

let $U = x^2+1$

$dU = 2x$. So, $\frac{1}{2}\int_0^2 2x(x^2+1)^3\,dx.$

$= \frac{1}{2}\int_0^2 U^3\,dU. \Rightarrow \frac{1}{2}\left[\frac{U^4}{4}\right]_0^2.$

$= \left[\frac{1}{8}(x^2+1)^4\right]_0^2$

$= \frac{1}{8}\left[(4+1)^4 - 0\right] \Rightarrow \frac{1}{8}(625)$

$= 78.12$ Ans

By Method 2:

let $x^2+1 = U$.

put $x=0$, $0^2+1 = U \Rightarrow 1$.

put $x=2$, $2^2+1 = U \Rightarrow 5$.

So,

$\frac{1}{2}\int_1^5 U^3\,dU \Rightarrow \frac{1}{2}\left[\frac{U^4}{4}\right]_1^5.$

$= \frac{1}{2}\left[\frac{625}{4} - \frac{1}{4}\right] \Rightarrow \frac{1}{2}\left(\frac{624}{4}\right) \Rightarrow \frac{624}{8} = 78$ Ans

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2nd Fundamental theorem

Dummy variable :-

used as a placeholder on limit.
used when there is indefinite limit. (∞).

e.g.

$$\int_2^x t^2\,dt \Rightarrow \left.\frac{t^3}{3}\right]_2^x \Rightarrow \frac{x^3}{3} - \frac{2}{3}$$

Ans will be as a function of that variable

2nd fund. Theorem:

(a) 1st integrate, then take derivative.

(b) $f(x) \rightarrow F(x) \rightarrow F'(x) \rightarrow f(x)$

we will get same function.

(c) $\frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x)$.

example:

$f(x) = x^3$. , $\int_1^x t^3\,dt$. chk validity!!

Sol

acc. to 2nd theorem,

$$\frac{d}{dx}\left[\int_1^x t^3\,dt\right] = f(x)$$

$$\frac{d}{dx}\left[\frac{t^4}{4}\right]_1^x \Rightarrow \frac{d}{dx}\left(\frac{x^4}{4} - \frac{1}{4}\right) \Rightarrow$$

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$$= \frac{d}{dx}\left[\frac{x^4}{4} - \frac{1}{4}\right]$$

$$= \frac{4x^3}{4}$$

$$x^3 = f(x).$$

hence proved !!

Applic. of Definite Integrals

Area Problems:

if $f(x) \geq g(x)$ for $a \leq x \leq b$.

then:

$$A = \int_a^b [f(x) - g(x)]\,dx.$$

Example:

Find area above by $y = x+6$

below by $y = x^2$. $[0, 2]$.

Sol:

$f(x) = x+6$, $g(x) = x^2$.

$$A = \int_0^2 [(x+6) - x^2]\,dx.$$

$$= \int_0^2 x\,dx + \int_0^2 6\,dx - \int_0^2 x^2\,dx.$$

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$$= \left[\frac{x^2}{2}\right]_0^2 + \left[6x\right]_0^2 - \left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{4}{2} + 12 - \frac{8}{3} \Rightarrow \frac{12+72-16}{6}$$

$$= \frac{68}{6} \Rightarrow \frac{34}{3} \text{ Ans.}$$

Example 2:

Find area enclosed b/w $y = x^2$ & $y = x+6$.

limit dii ee nhi! 😊

compare both:

$$x^2 = x+6$$

$$x^2 - x - 6 = 0$$

$$x^2 + 2x - 3x - 6 = 0$$

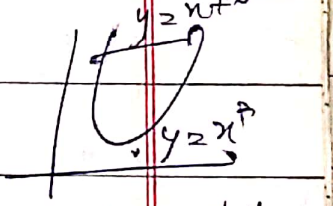
$$x(x+2) - 3(x+2) = 0$$

$$(x+2)(x-3) = 0$$

$$\boxed{x = -2} \quad \boxed{x = 3}.$$

$y = x^2$ → parabola,

$y = x+6$ → line



that's why,

$(x+6) - (x^2)$

Now, for Area:

$$A = \int_{-2}^{3} \left([x+6] - x^2\right) dx$$

$$= \left[\frac{x^2}{2} + 6x - \frac{x^3}{3}\right]_{-2}^{3}$$

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$$= \left(\frac{3^2}{2} + 6(3) - \frac{3^3}{3}\right) - \left(\frac{(-2)^2}{2} + 6(-2) - \frac{(-2)^3}{3}\right)$$

$$= \left(\frac{9}{2} + 18 - 9\right) - \left(2 - 12 + \frac{8}{3}\right)$$

$$\left(\frac{9+36-18}{2}\right) - \left(\frac{6-36+8}{3}\right)$$

$$\frac{27}{2} + \frac{22}{3}$$

$$\frac{81+44}{6} \Rightarrow \frac{125}{6} \text{ Ans}$$

Volume by Slicing

→ volume of cylinder $V = A \cdot h$.

for 2D.

But for 3D $V = ?$

→ we apply method of slicing on a 3D-shaped volume.

① width would be Δx_k.

② height would be x_k^*.

volume of 3D would be $A \times h \times w$.

③ $V_k \approx A(x_k^*)\,\Delta x_k$

for all slices, sum up:

$$V = \sum_{i=1}^{n} A(x_k^*)\,\Delta x_k$$

take $\lim_{\max \Delta x_k \to 0}$ i.e., $\lim_{\max \Delta x_k \to 0} \sum_{i=1}^{n} A(x_k^*)\Delta x_k$

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we can write it in integral form

$$V = \int_a^b A(x)\,dx.$$

roughly, volume of cylinder is obtained by integrating its area b.b.

Revolution about axis formulas:

1- Disks:

$A = \pi r^2$. so, $r = f(x)$.

then

$$V = \int_a^b \pi [f(x)]^2\,dx$$

2- washers:

It contain 2 functions. so,

$$A = \pi [f(x)]^2 - \pi [g(x)]^2$$
$$= \pi [f(x)^2 - g(x)^2].$$

so,

$$V = \int_a^b \pi [[f(x))^2 - [g(x)^2]]\,dx$$

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Example:

Derive formula for volume of sphere.

remember:

$$x^2 + y^2 = r^2.$$

$$y = f(x) = \sqrt{r^2 - x^2}.$$

$a = -r$
$b = r$] intervals on -ve & +ve x-axis.

$$V = \int_{-r}^{r} \pi \left[\sqrt{r^2 - x^2}\right]^2 dx.$$

$$= \int_{-r}^{r} \pi (r^2 - x^2)\, dx.$$

$$\therefore \pi \left[xr^2 - \frac{x^3}{3} \right]_{-r}^{r}$$

$$= \pi \left[\left(r^3 - \frac{r^3}{3} \right) - \left(-r^3 + \frac{r^3}{3} \right) \right].$$

$$= \pi \left[-r^3 - \frac{r^3}{3} + r^3 - \frac{r^3}{3} \right].$$

$$= \pi \left[2r^3 - 2\frac{r^3}{3} \right].$$

$$= \pi \left[\frac{6r^3 - 2r^3}{3} \right] \Rightarrow \pi \left(\frac{4r^3}{3} \right).$$

$$\boxed{V = \frac{4\pi r^3}{3}} \quad \text{(volume of sphere)}$$

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Volume of cylindrical Shells !!!

when we extend a washer upward, we generate a solid.

its volume would be:

$V = (\text{area of cross-section}) \cdot (\text{height}) = (\pi r_2^2 - \pi r_1^2) h$

$= \pi (r_2^2 - r_1^2) h$

$= \pi (r_2 - r_1)(r_2 + r_1) h.$

$V = 2\pi \cdot \left[\frac{1}{2}(r_2 + r_1)\right] \cdot r_2 - r_1 \cdot h \rightarrow$ height

avg. radius, thickness.

if we put:

average radius $= x_k^*$

thickness $= \Delta x.$

height $= f(x_k^*)$. our volume will be

$V = 2\pi x_k^* f(x_k^*) \Delta x_k$

Now (wohi kaam)

$\sum_{k=1}^{n} 2\pi x_k^* f(x_k^*) \Delta x_k$

$\lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} 2\pi x_k^* f(x_k^*) \Delta x_k$

Final formula

$$\int_a^b 2\pi x f(x) dx$$

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Example: find volume of solid b/w

$y = \sqrt{x}$, $x = 1$, $x = 4$.

x axis revolved about y.

$$V = \int_1^4 2\pi x \sqrt{x}\, dx.$$

$$= 2\pi \int_1^4 x^{\frac{3}{2}}\, dx.$$

$$= 2\pi \left[\frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1} \right]_1^4$$

$$= \left(2\pi \cdot \frac{2}{5} \left(x^{\frac{5}{2}} \right) \right)_1^4$$

$$= \frac{4\pi}{5} \left[(4)^{\frac{5}{2}} - 1^{\frac{5}{2}} \right].$$

$$= \frac{4\pi}{5} (32 - 1)$$

$$= \frac{128\pi}{5} \text{ Ans.}$$

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Length of plane Curve

اگر کوئی function کسی interval $[a,b]$ پر continuous ہو تو اسے Graph کے لحاظ سے smooth function کہتے ہیں۔ ⇐

⇒ smooth curves کو

⇒ general form of line is curve ⇐

⇒ arc is a collection of arch ⇐

Pythagoras theorem is used in it.

$$c^2 = b^2 + a^2$$

$$L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

by Substitution w.r.t MVT.;

$$L_k = \sqrt{1 + (f'(x_k^*))^2}\ \Delta x_k$$

apply summation & $\Delta x_k \to 0$ as max.

so,

$$L = \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} \sqrt{1 + (f'(x_k^*))^2}\ \Delta x_k$$

finally

$$\boxed{L = \int_a^b \sqrt{1 + (f'(x))^2}\, dx}$$

Arc length. formula.

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Surface area of revolution

Formula for surface area of frustum:

$$S = \pi (r_1 + r_2) l$$

final formula of surface area:

$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2}\, dx.$$

Example:

Find surface area of sphere bla bla bla...

$y = \sqrt{1 - x^2}$ $\quad 0 \le x \le \frac{1}{2}$ above x-axis.

Solution:

let $y = f(x)$

$f'(x) = \frac{d}{dx}(1 - x^2)^{\frac{1}{2}}$

$f'(x) = \frac{1}{2}(1 - x^2)^{-\frac{1}{2}}(-2x)$

put in formula:

$$S = \int_0^{\frac{1}{2}} 2\pi \sqrt{1 - x^2} \sqrt{1 + \left(\frac{-x}{\sqrt{1 - x^2}}\right)^2}\, dx.$$

$$= \int_0^{\frac{1}{2}} 2\pi \sqrt{1 - x^2} \sqrt{1 + \frac{x^2}{1 - x^2}}\, dx.$$

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$$= \int_0^{\frac{1}{2}} 2\pi \sqrt{1-x^2} \sqrt{\frac{1-x^2+x^2}{1-x^2}}\, dx$$

$$= \int_0^{\frac{1}{2}} 2\pi \sqrt{1-x^2} \cdot \frac{1}{\sqrt{1-x^2}}\, dx$$

$$= \int_0^{\frac{1}{2}} 2\pi\, dx$$

$$= 2\pi \int_0^{\frac{1}{2}} dx.$$

$$= 2\pi [x]_0^{1/2}$$

$$= 2\pi \left(\frac{1}{2} - 0\right).$$

$$= \frac{2\pi}{2} \Rightarrow \pi \quad \text{Ans.}$$

work & definite integral

1- **Work done by a constant force:**

$W = F.d$ = force x distance.

units: Force = pound, Newton, Dynes

Work = footpound, Newton Meter, Dyne cm

1 Nm = 1 Joule.

Dyne for $gm\,cm\,s^{-2}$

Newton for $kg\,m\,s^{-2}$

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work done by a variable force = definite integral -

In term of definite;

force = $f(x_k^*)$.

distance = Δx_k.

$W = F(x_k^*)\Delta x_k$ -

$= \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} f(x_k^*)\Delta x_k$.

$$W = \int_a^b f(x)\,dx.$$

density of water = 62.4. = ρ.

pressure = F/A

$= \rho h$.

Fluid pressure

$F = \rho h A$

⇒ Force does not depend on shape of container

Pascal Principle

Fluid pressure is same in all directions at a given height.

Fluid pressure formula

$$F = \int_a^b \rho h(x)\, w(x)\, dx$$

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Improper Integral

① Definite integral has finite limits which represents area under the curve.

② Improper integral has endpoint goes to infinity

$$\int_a^{+\infty} f(x)\,dx = \lim_{l \to +\infty} \int_a^{l} f(x)\,dx.$$

infinite upper limit ki jaga letter likhna h. or uski limit ∞ le leni--- waooo!

=

agar limit exist ki to Integral converge
agar limit does not exist to phr Integral Diverge

converge means gradually 0 ko approach kry-

Example:

Evaluate $\int_1^{+\infty} \frac{1}{x^2}\,dx$

$$\lim_{l \to \infty} \int_1^{l} \frac{1}{x^2}\,dx.$$

$$\lim_{l \to \infty} \left[-\frac{1}{x}\right]_1^{l} \Rightarrow -\frac{1}{l} - (-1) \Rightarrow 1 - \frac{1}{l}$$

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Now if $l \to \infty$.

$$\int_1^{+\infty} \frac{dx}{x^2} = \lim_{l \to \infty} \left(1 - \frac{1}{l}\right).$$

$$= 1 - 0$$

$$= 1 \quad \text{Ans.}$$

E.g 2

$$\int_1^{+\infty} \frac{dx}{x}$$

$$= \int_1^{+\infty} \frac{1}{x}\, dx.$$

$$= \lim_{l \to \infty} \int_1^{l} \frac{1}{x}\, dx.$$

$$= \lim_{l \to \infty} [\ln |x|]_1^l$$

$$= \lim_{l \to \infty} \ln l - \ln 1$$

$$= \lim_{l \to \infty} \ln l - 0.$$

$$= \infty \quad \text{Ans}$$

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e.g 3: $\int_1^4 \frac{dx}{(x-2)^{2/3}}$

Sol:

$$\int_1^4 \frac{1}{(x-2)^{\frac{2}{3}}} = \int_1^2 \frac{1}{(x-2)^{\frac{2}{3}}} dx + \int_2^4 \frac{1}{(x-2)^{\frac{2}{3}}} dx.$$

$$\lim_{l \to 2^+} \int_1^l \frac{1 \cdot dx}{(x-2)^{\frac{2}{3}}} = \int_1^l (x-2)^{-\frac{2}{3}} dx$$

$$= 3 \int_1^l (x-2)^{\frac{1}{3}} dx \Rightarrow 3\left[(l-2)^{\frac{1}{3}} - (1-2)^{\frac{1}{3}}\right]$$

$$= 3.$$

$$\lim_{l \to 2^-} \int_2^4 (x-2)^{-\frac{2}{3}} dx$$

---- Same as #1

L' Hopital's Rule

converge means finite value aygi

Diverge means ~~infinite~~ value ayegi.

L'Hopital's Rule kehta ha:

when we take derivative of both functions. & then take limit of that function —— Then majically answer me infinity nhi hogi —— xD

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Example:

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}.$$

limit direct put Nhi krskty.

Then, use L'Hopital Rule:

$$\lim_{x \to 2} \frac{x^2-4}{x-2} \Rightarrow \lim_{x \to 2} \frac{\frac{d}{dx}(x^2-4)}{\frac{d}{dx}(x-2)}$$

$$= \lim_{x \to 2} \frac{2x}{1}$$

$$= 2(2) = 4 \text{ Ans.}$$

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